



Assignment 01

E-Commerce Information System Investigation Information System Analysis and Design

A01

ABC Fashion
Online Shop

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Executive Summary

ABC Fashion Online Shop is modeled as a medium-sized B2C retailer that owns and sells apparel, footwear, and accessories online. The investigation identifies one focal problem: fragmented and unreliable information prevents the business from coordinating Product Variants, stock, Orders, payment results, delivery, returns, customers, and reporting as one end-to-end process. In fashion, the stocked and fulfilled unit is the exact size/color **Product Variant**, not its parent Product Style.

The proposed **ABC Fashion Commerce Information System (AFCIS)** establishes a controlled information backbone for catalog, customer, cart, Order, promotion, variant-level inventory, fulfillment, after-sales, review, and operational reporting. Payment execution, carrier operations, and message delivery remain external services. Six problems are established by the assignment case; return/exchange and promotion-control concerns remain hypotheses for validation before Assignment 02.

1. Introduction

This report answers the assignment's central question: *What problem should the information system solve?* It follows the required progression from organization and activities through stakeholders, information, problems, scope, boundary, and objectives. It is a desk-based investigation grounded in the Assignment 01 case and the completed fashion-domain analysis; detailed UML, database, interface, and implementation design are intentionally deferred.

Assumptions and limits. ABC is treated as one medium-sized, owned-inventory B2C retailer with one initial warehouse and an online storefront. Payment authorization, physical transport, and message delivery are performed by external providers. The case supplies no interviews, logs, policies, or measured baselines; therefore, fashion-specific failure modes and numeric targets are explicitly preliminary.

Controlled terms. A **Product Style** groups related merchandise; a **Product Variant** is one sellable size/color/attribute combination with one SKU or GTIN. Variant attributes and identifiers must distinguish the precise sellable item [R2], [R3]. It is the stock, order-line, fulfillment, return, and exchange unit. **Available-to-Sell (ATS)** is eligible on-hand stock less reservations and approved safety stock.

2. Business Background

2.1. Organization

ABC Fashion Online Shop selects, purchases, markets, sells, fulfills, and supports its own assortment of apparel, footwear, and accessories through online channels. Its purpose is to convert a managed assortment into reliable orders while supplying enough size, fit, color, price, availability, delivery, and return information for confident customer decisions.

2.2. Business Model

Aspect	Investigation finding
Customer/value	Online consumers; convenient discovery, trustworthy variant availability, and transparent order/after-sales status.
Revenue/channel	Owned-merchandise sales through a responsive online storefront, net of promotions, returns, refunds, and fulfillment costs.
Relationships	Self-service browsing, cart, checkout, tracking, reviews, returns/exchanges, and staff-assisted support.
Resources and partners	Product and stock information, staff, warehouse, suppliers, Payment Gateway, Shipping Provider, and Notification Service.

2.3. Products and Customers

The assortment covers apparel, footwear, and accessories. Customers need search and filtering by category, price, size, color, material, fit, and availability; variant-consistent media and size guidance; accurate promotion and ATS information; and dependable payment, shipment, return, exchange, and refund status. Apparel UX research particularly supports clear fit and size guidance for confident product selection [R4]. A style can be browsed as one family, but selection must resolve to a unique SKU before checkout.

2.4. Major Business Activities

ABC manages assortment and variants; prices/promotions; customers and consent; browsing/search; cart/checkout; orders and payments; variant-level stock; picking, packing, shipping and tracking; customer service, cancellations, returns/exchanges/refunds; reviews; supplier receipts; and sales/stock/operations reporting.

Business objectives: trustworthy product presentation; fulfillment of the exact promised variant; less re-keying and disconnected handoffs; timely self-service status and after-sales support; and timely, reconcilable management information.

Evidence discipline. Inconsistent products, outdated inventory, manual order work, unclear order status, fragmented customer records, manual reports, and limited real-time information are case facts. Specific variant, returns, and promotion failure modes are hypotheses until validated.

3. Stakeholder Analysis

Internal/external classification is relative to ABC as an organization. Employees later appear as external entities in the context DFD because they stand outside the AFCIS software boundary.

3.1. Internal Stakeholders

Owner/General Manager; E-Commerce Administrator; Merchandiser/Catalog Staff; Marketing/Promotion Staff; Sales and Customer Service Staff; Warehouse/Fulfillment Staff; and Finance/Accounting Staff.

3.2. External Stakeholders

Customer; Supplier; Payment Gateway; Shipping Provider; and Notification Service.

3.3. Stakeholder Interests

Stakeholder	Role	Main interest / need	Influence
Owner / Manager	Sponsor and policy owner	Benefits, risks, reliable KPIs and timely exceptions	High
Administrator	Access, settings, exceptions	Efficient, accurate and auditable control	High
Merchandiser / Catalog Staff	Catalog and product data	One controlled Style/Variant source with complete attributes	High
Marketing / Promotion Staff	Campaigns and promotion rules	Deterministic eligibility, dates, stacking and reporting	High
Sales / Customer Service	Order and after-sales support	Consolidated customer/order lifecycle status	High
Warehouse / Fulfillment	Stock and physical events	Exact-SKU work, accurate balances and traceability	High
Finance / Accounting	Reconciliation	Linked order, payment, refund and fee information	Med-high
Customer	Browse, buy, track, return, review	Confidence, accurate availability, fair price and status	High interest
Supplier	Provide merchandise/facts	Clear receipt and replenishment communication	Medium
Payment Gateway	Financial authorization, capture and refund	Valid requests, stable references and reconcilable outcomes	High interface
Shipping Provider	Shipment and tracking services	Complete delivery data, milestones and exception handling	High interface
Notification Service	Message delivery	Valid destination/content and correlation reference	Medium

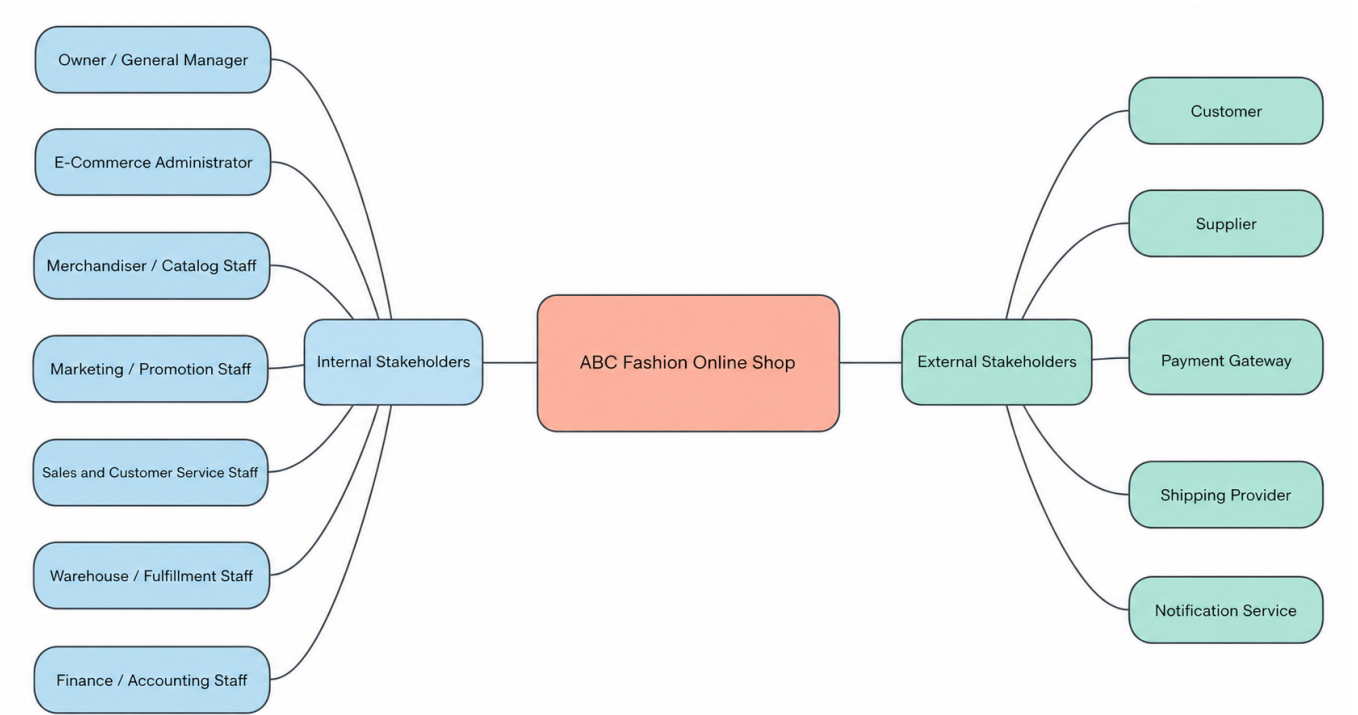


Figure 1: Core Diagram 1 — Stakeholder Map for ABC Fashion Online Shop.

Figure 1 classifies the internal and external stakeholders relative to ABC's organizational boundary. The map contains all and only the stakeholder groups analyzed above. Branch connectors indicate classification, not authorization or data flow; therefore, UML actors and Use Cases are not used.

4. Business Activity Analysis

Required activity	Role in the business	Principal information
4.1 Product Management	Maintains categories, Styles, Variants, images, size/fit/care, lifecycle, prices and publication.	Style/Variant, media, price, promotion, supplier facts
4.2 Customer Management	Maintains one controlled identity, contact, address, consent and service timeline.	Customer, consent, address, service history
4.3 Order Processing	Validates cart, price, promotion and ATS; snapshots exact SKU lines; reserves stock and controls status.	Cart, Order, Order Line, reservation, status event
4.4 Payment	Sends authorization/capture/refund requests to the gateway and records results/references.	Amount, request, gateway reference/status; no raw PAN/CVV [R5]
4.5 Inventory	Records SKU/location receipts, reservations, releases, picks, shipments, returns and adjustments.	On-hand, reserved, ATS, movement and source reference
4.6 Shipping	Directs exact-SKU pick/pack, requests shipment, and maps tracking milestones into lifecycle status.	Pick list, package, address, tracking/status
4.7 Reporting	Produces defined sales, stock, fulfillment, return/refund, promotion and exception views.	Governed operational data, period/status filters, refresh time

Browsing and fit support, cart and checkout, promotions, reviews, cancellation, return, exchange, refund, and supplier receipt complete the required fashion journey. Figure 2 presents the high-level order-to-after-sales business process and its external service interactions.

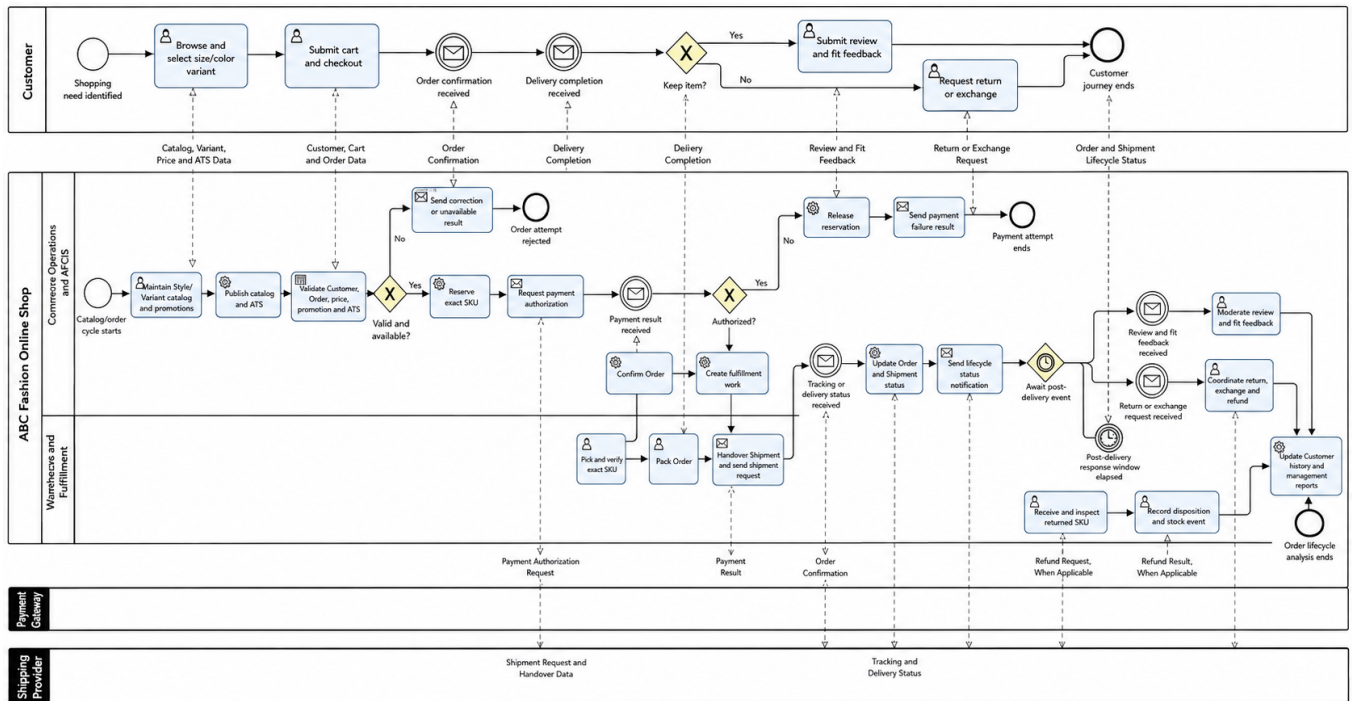


Figure 2. Core Diagram 2 - High-Level Fashion Order-to-After-Sales Business Process.

Solid Sequence Flows remain inside a Pool; dashed Message Flows cross participant boundaries, as required by BPMN 2.0 and the Visual Paradigm notation guidance [R6], [R7]. External providers are black-box Pools. Alternative validation, payment-failure, review, and return/exchange paths are shown without introducing detailed Use Cases.

5. Information Analysis

5.1. Information Produced

Approved catalog/variant records, carts, Orders and line snapshots, reservations/movements, payment/refund requests, confirmations, pick/pack work, shipment requests/status, notifications, after-sales cases, reviews, audit events and reports.

5.2. Information Consumed

Supplier product facts; customer identity, contact, address and consent; searches, carts and checkout selections; size and fit data; price and promotion rules; ATS; gateway results; warehouse events; carrier milestones; return reasons; and report parameters.

5.3. Information Stored

AFCIS stores controlled Product Style/Variant masters; media/size-fit metadata; prices/promotions; customer data; carts as policy permits; Orders and immutable line snapshots; SKU inventory/reservations/movements; gateway references/statuses; fulfillment, shipment and notification events; returns/exchanges/refunds; reviews and audit history. Raw card numbers/CVV, carrier routing, supplier manufacturing data and the accounting ledger remain outside.

5.4. Information Catalogue

ID	Information	Source	Purpose / principal users	Lifecycle
I-01–03	Style, Variant/SKU, media and size/fit/care	Merchandiser; supplier facts	Exact product discovery, selection, stock and fulfillment	P/C/S/X
I-04	Price and promotion rule/result	Marketing / Manager	Consistent presentation, checkout and reporting	P/C/S/X
I-05	Customer, address, preference and consent	Customer / authorized staff	Checkout, service and controlled communication	P/C/S/X
I-06–07	Cart, checkout, Order and line snapshot	Customer / AFCIS	Commercial commitment and coordination	P/C/S/X
I-08	Inventory balance, reservation, movement, ATS	Warehouse / AFCIS	Variant-level availability and traceability	P/C/S
I-09	Payment/refund request, reference and status	AFCIS / Gateway	Coordinate and reconcile without raw card data	P/C/S/X
I-10–12	Fulfillment, shipment/tracking, notification	Warehouse / providers	Direct work and explain lifecycle status	P/C/S/X
I-13	Cancellation, return, exchange and refund case	Customer, staff, Gateway	Control reverse logistics, replacement and money	P/C/S/X
I-14	Review and fit feedback	Customer / moderator	Product insight and approved social proof	P/C/S/X
I-15	Supplier, receipt and discrepancy	Supplier / Warehouse	Replenishment and stock receipt administration	P/C/S
I-16–17	KPI/report and status/audit event	AFCIS / actors	Decisions, exceptions, reproducibility, accountability	P/C/S/X

P = produced/updated; C = consumed; S = stored; X = exchanged across the AFCIS boundary.

Preliminary information-quality rules

1. Every active Product Variant belongs to one Product Style and has a unique SKU; the Style itself is not stocked or ordered.
2. Published Variants contain applicable size/system, color, material/pattern/fit, approved media, price, state and availability source.
3. Every stock event references exact SKU, location, signed quantity, event type/time, actor/source and originating reference.
4. An Order Line preserves the committed SKU, description, attributes, price, discount and quantity after catalog changes.
5. Order, Payment, Shipment, Return, Exchange and Refund use controlled, timestamped status transitions.
6. Payment data is minimized; raw PAN and CVV/CVC are excluded [R5]. Customer/consent changes are attributable.
7. Reports state period, included statuses and refresh/cut-off time so results are reproducible.

All major activities have identifiable inputs and outputs, and the Product Variant remains the cart, stock, order-line, pick, return and exchange unit throughout.

6. Current Problems

6.1. Problem Identification

ID	Significant problem	Affected information	Evidence
P-01	Inconsistent product information can make Style, size/color Variant, media, price and publication disagree.	I-01-04	Case fact; fashion extension
P-02	Outdated inventory may not represent ATS for the exact size/color SKU.	I-02, I-08	Case fact; variant extension
P-03	Manual order steps and disconnected handoffs cause delay and re-keying risk.	I-06-11	Case fact
P-04	Customers/staff cannot always obtain one accurate, timely lifecycle status.	I-07, I-09, I-11-13, I-17	Case fact; after-sales extension
P-05	Customer data in different places creates duplicate/inconsistent identity, address, consent and history.	I-05, I-07, I-12-13	Case fact
P-06	Manual reports and limited real-time information delay and weaken management decisions.	I-07-09, I-11, I-13, I-16	Case fact
P-07	Return/exchange, inspection, restock and refund may lack one traceable lifecycle.	I-02, I-07-09, I-13, I-17	Domain hypothesis
P-08	Promotion rules may be inconsistent across catalog, cart, order, service and reports.	I-04, I-06-07, I-16-17	Hypothesis

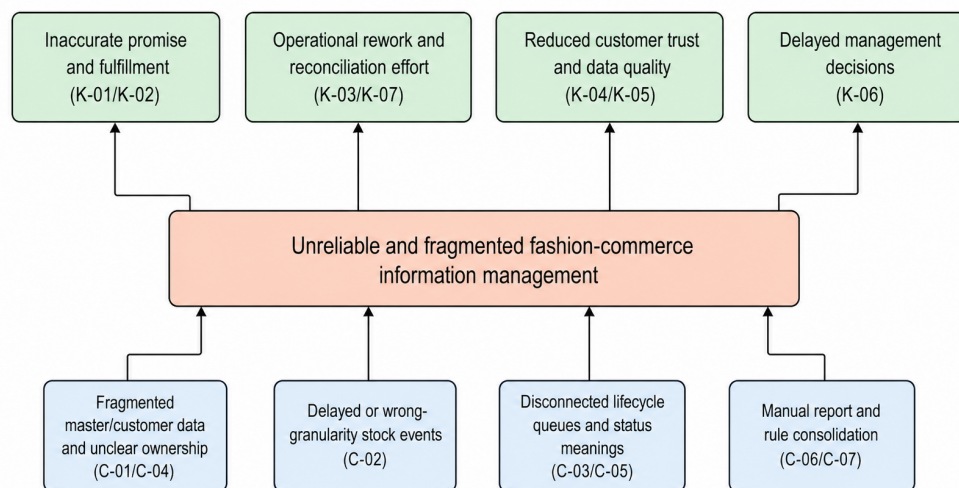
6.2. Problem Causes

Candidate causes are: no governed Style/Variant master or clear ownership (C-01); delayed/manual or wrong-granularity stock events (C-02); separate order/payment/stock/shipping/after-sales queues (C-03); fragmented customer capture (C-04); inconsistent status vocabulary/ownership (C-05); manual report consolidation (C-06); and uncontrolled promotion eligibility/effective-date/stacking rules (C-07). These require catalog audits, process walkthroughs, cycle counts, source inventories and status/rule workshops.

6.3. Problem Consequences

The combined effects are inaccurate promise or cancellation; overselling, underselling and wrong-variant picks; staff re-keying and reconciliation; delayed or conflicting customer status; duplicate customer and consent risk; slower assortment, replenishment and promotion decisions; and broken financial and operational traceability.

6.4. Problem Tree



Note: Root causes are candidate causes pending stakeholder interviews and operational evidence.

PT-1. Mandatory supporting Problem Tree. Causes appear below the focal problem and consequences above it; arrows mean “contributes to.”

PT-1 presents the candidate causal relationships among root causes, the focal problem, and business consequences.

P-01–P-04 are investigated first because they determine whether the exact Variant can be promised, paid, fulfilled and explained. P-05–P-06 share the same source-of-truth need; P-07–P-08 remain in preliminary scope only if early validation confirms them.

7. Proposed Information System

7.1. System Purpose

The proposed **ABC Fashion Commerce Information System (AFCIS)** provides one controlled operational backbone for Style/Variant, customer, price/promotion, Order, payment-reference, inventory, fulfillment, shipment, after-sales, review and reporting information. It is justified by P-01–P-06, not by a solution-first desire to build a website.

7.2. System Scope

ID	Capability inside AFCIS	Problems addressed
S-01	Controlled category, Style/Variant, media, size/fit/care and publication management	P-01, P-02
S-02	Customer profile, address, consent and authorized service-history management	P-04, P-05
S-03	Variant-aware browsing, cart, checkout, price/promotion and ATS validation	P-01, P-02, P-08
S-04	Order snapshot/status, reservation, cancellation, payment/refund orchestration and confirmation	P-02–04, P-07–08
S-05	SKU/location stock, reservation, movement, pick/pack, adjustment and return disposition	P-01–03, P-07
S-06	Shipment request/tracking integration, lifecycle timeline and notification orchestration	P-03, P-04
S-07	Return, exchange, inspection, replacement and refund-case coordination	P-04, P-07
S-08	Eligible review, moderation state and fit-feedback capture	Supports business objectives
S-09	Operational dashboards/reports, exceptions, reconciliation and audit history	P-02–04, P-06, P-08
S-10	Roles, reference values, status definitions and workflow/policy administration	P-01, P-04–06, P-08

Scope principles are one operational source of truth, exact-Variant traceability, event-linked lifecycle status, and external specialization remaining external. The scope defines information responsibility, not screens, tables, APIs, hosting or code.

7.3. System Boundary

Area	Inside AFCIS	Outside / boundary rationale
Customer and product	Catalog, account, cart, checkout, Order, status, content, price/promotion	Customer device/decisions; supplier manufacturing and creative tools
Payment	Request orchestration, amount/reference/status and reconciliation	Gateway captures credentials and executes authorization/capture/refund
Inventory and warehouse	Stock records, ATS, reservations, work queues and physical-event capture	Merchandise, shelving/equipment and human physical execution
Shipping and notification	Shipment/message requests, tracking/status, customer timeline and logs	Carrier routing/transport and provider delivery infrastructure
Service and reporting	Consolidated timeline, cases, KPIs, exceptions and governed exports	Human conversations/discretion; general ledger, tax, payroll and enterprise BI
Procurement	Supplier master, expected receipt, discrepancy and replenishment note	Supplier portal, sourcing, contracts and full purchase-to-pay

Explicit exclusions: detailed UML Use Cases and requirements; ERD/database/component/deployment/API/UI design and code; marketplace sellers/commissions; physical POS; payment credentials and gateway internals; carrier fleet/routing; supplier manufacturing/procurement automation; general ledger/tax/payroll/HR; and advanced AI recommendation, dynamic pricing or forecasting.

Dependencies and controls: provider contracts/interfaces and exception rules; catalog/customer/stock/order profiling before migration; approved publication, promotion, reservation, cancellation, return/exchange/refund and status policies; role-based handling of personal/payment-restricted information; and timely staff capture of physical events.

Boundary rationale. AFCIS records and explains requests/results but does not perform payment processing, parcel transport or message delivery. Supplier is a business stakeholder but not a direct system interface in the initial boundary; authorized staff record relevant receipt/replenishment inputs.

7.4. Context Diagram

Figure 3 uses one black-box DFD process, seven external entities, sixteen directed noun-phrase information flows and no data store or internal subprocess, consistent with Visual Paradigm's DFD guidance [R8].

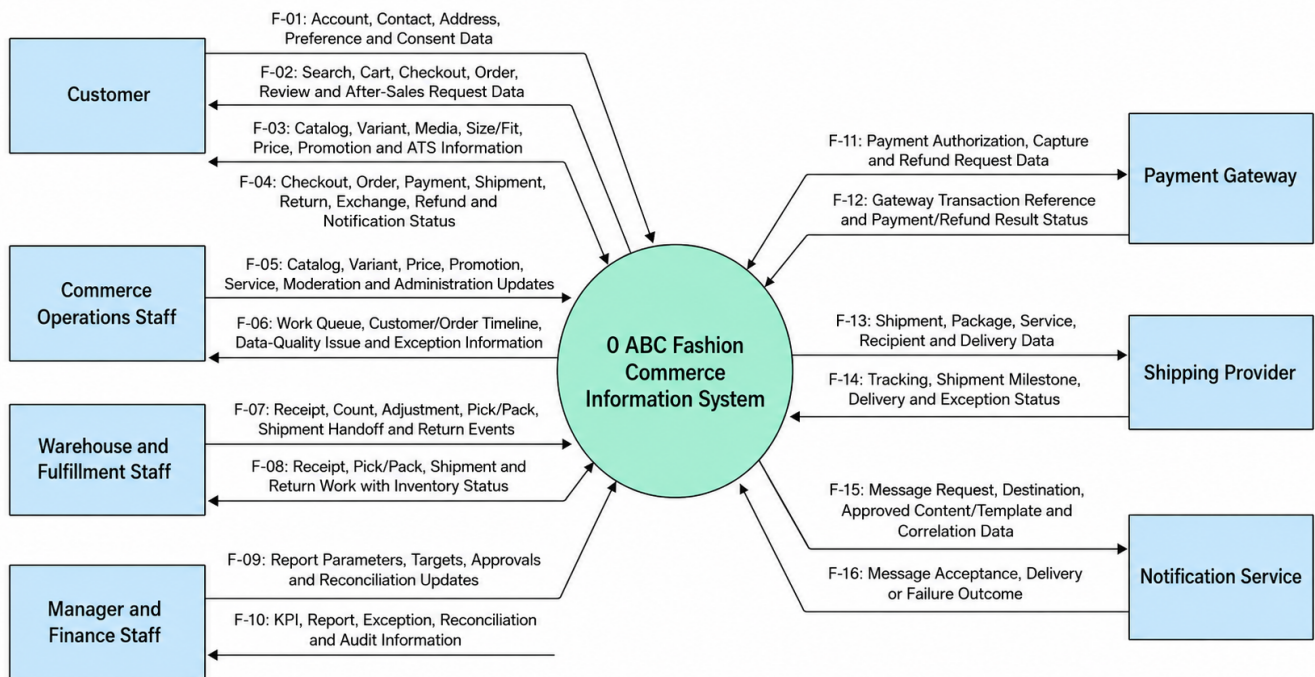


Figure 3: Core Diagram 3 — AFCIS System Context Diagram (Yourdon–DeMarco DFD convention).

7.5. Information Flows

Flow	Source	Destination	Information crossing the boundary
F-01	Customer	AFCIS	Account, Contact, Address, Preference and Consent Data
F-02	Customer	AFCIS	Search, Cart, Checkout, Order, Review and After-Sales Request Data
F-03	AFCIS	Customer	Catalog, Variant, Media, Size/Fit, Price, Promotion and ATS Information
F-04	AFCIS	Customer	Checkout, Order, Payment, Shipment, Return, Exchange, Refund and Notification Status
F-05	Commerce Operations Staff	AFCIS	Catalog, Variant, Price, Promotion, Service, Moderation and Administration Updates
F-06	AFCIS	Commerce Operations Staff	Work Queue, Customer/Order Timeline, Data-Quality Issue and Exception Information
F-07	Warehouse/Fulfillment Staff	AFCIS	Receipt, Count, Adjustment, Pick/Pack, Shipment Handoff and Return Events
F-08	AFCIS	Warehouse/Fulfillment Staff	Receipt, Pick/Pack, Shipment and Return Work with Inventory Status
F-09	Manager/Finance Staff	AFCIS	Report Parameters, Targets, Approvals and Reconciliation Updates
F-10	AFCIS	Manager/Finance Staff	KPI, Report, Exception, Reconciliation and Audit Information
F-11	AFCIS	Payment Gateway	Payment Authorization, Capture and Refund Request Data
F-12	Payment Gateway	AFCIS	Gateway Transaction Reference and Payment/Refund Result Status
F-13	AFCIS	Shipping Provider	Shipment, Package, Service, Recipient and Delivery Data
F-14	Shipping Provider	AFCIS	Tracking, Shipment Milestone, Delivery and Exception Status
F-15	AFCIS	Notification Service	Message Request, Destination, Approved Content/Template and Correlation Data
F-16	Notification Service	AFCIS	Message Acceptance, Delivery or Failure Outcome

Every context entity has at least one named incoming and outgoing flow. Payment, shipping and notification providers remain external. The DFD contains no entity-to-entity flow, database, UI or implementation component.

7.6. System Objectives

ID	Preliminary objective	Candidate measure / target
O-01	One governed Style/Variant catalog with complete fashion attributes and controlled publication.	100% of published Variants pass mandatory-field and unique-SKU validation.
O-02	Timely, traceable SKU-level inventory, reservations, movements and ATS.	At least 99% physical-to-record accuracy; accepted event visible within 60 seconds.
O-03	Coordinate routine Orders from checkout through payment, stock, fulfillment and shipment without re-keying.	At least 90% of routine paid Orders enter warehouse work without manual re-entry.
O-04	One timely, explainable lifecycle status for customers and staff with proactive notifications.	At least 95% of accepted status events visible within five minutes.
O-05	Traceable cancellation, return, size/color exchange, inspection, restock, replacement and refund.	100% of approved cases retain linked Order Line/SKU, decision, disposition and financial status.
O-06	Controlled customer data and secure external payment coordination without raw card credentials.	100% of AFCIS card-payment records use gateway references; zero raw PAN/CVV is stored; duplicate-customer and consent-quality targets are established after authorized data profiling.
O-07	Timely, reproducible operational reports and exception views.	Agreed dashboards refresh within 15 minutes; standard reports need no manual source consolidation.

Targets are provisional until ABC measures the current baseline and approves definitions in Assignment 02. Together, O-01–O-07 trace to every case-established problem P-01–P-06; O-02/O-05 and O-01/O-07 also cover P-07/P-08 if those hypotheses are confirmed.

8. Conclusion

ABC does not primarily need “a website.” It needs to solve a more fundamental information problem: fragmented and unreliable Style/Variant, inventory, customer, Order, external-service status, after-sales and reporting information prevents the business from promising, fulfilling, explaining and learning from the lifecycle of the exact fashion Product Variant.

AFCIS responds with a bounded, traceable information-system proposal. It governs the Style/Variant distinction and coordinates catalog, customer, cart, Order, payment-reference, stock, fulfillment, shipment, return/exchange/refund, review and reporting information. Payment execution, physical transport, message delivery, supplier systems and accounting stay outside with named crossings. Before Assignment 02, ABC should validate candidate causes and P-07/P-08, baseline objective measures, confirm policies/status vocabularies, and verify provider interfaces. Only then should stakeholders, problems, activities and information flows become formal actors, requirements and Use Cases.

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